

Pressure Shadows, Strain Heterogeneity & Pressure Solution Microstructures

Authoritative References: John D. Winter (Ch 23: Structures & Textures) & Van der Pluijm & Marshak (Ch 9 & 11: Pressure Solution & Cleavage)

Comprehensive Chapter Masterclass: Metamorphic Textures & Stress Heterogeneity

Rigid Clasts in Ductile Matrices, Pressure Solution Transfer, Fiber Growth & Strain Shadows

1. Stress Shielding & Generation of Pressure Shadows Around Rigid Inclusions

When a deforming rock contains heterogeneous components—specifically **rigid grains or porphyroblasts** (such as pyrite, garnet, magnetite, or quartz pebbles) embedded within a more ductile, flow-prone matrix (such as calcite, phyllosilicates, or fine-grained quartz)—the local stress field is severely perturbed:

- **Stress Concentration:** On grain boundaries perpendicular to the maximum compressive stress (σ_1), compressive stress reaches peak intensity. The ductile matrix impinges against the rigid barrier, creating zones of extreme mean stress.
- **Stress Shielding:** Adjacent to the flanks of the rigid clast (in the extension direction σ_3 , or parallel to foliation), the rigid crystal supports the compressive load, mechanically shielding the matrix behind it. This creates **local zones of significantly reduced mean stress (low pressure)**.

"Pre-kinematic crystals, when clearly older than the deformation, commonly exhibit microstructures resulting from high localized strain... Pressure shadows (Figure 34c) occur when solution transfer dissolves a mineral (usually a matrix mineral) from high-stress areas and precipitates it in relatively low-stress areas adjacent to a porphyroblast. In addition, the external foliation usually wraps around a pre-kinematic porphyroblast, a result of flattening in the matrix."

— John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2nd Ed., Chapter 23, p. 527)



FIGURE 34 Typical textures of pre-kinematic crystals. (a) Bent crystal with undulose extinction. (b) Foliation wrapped around a porphyroblast. (c) Pressure shadow or fringe. (d) Kink bands or folds. (e) Fragmented and stretched crystals (microboudinage). (f) Deformation twins. From Spry (1969).

Post-kinematic crystallization either outlasted deformation or occurred in a distinct later thermal or contact event. When clearly distinguishable, both the previous deformation and the later mineral growth or recrystallization must be apparent. Post-kinematic growth results in unstrained crystals. In many cases, the crystals are randomly oriented and cut across an earlier foliation (Figure 35b), although mimetic growth may cause a large proportion of later crystals to grow parallel to the foliation anyway.

Pseudomorphs (Figure 35f), particularly those in which a crystal is replaced by an aggregate of random smaller crystals, suggest that the replacement was post-kinematic. Bent crystals may polygonize during later static growth to smaller unstrained crystals. A dramatic example of this is the **polygonal arc** (Figure 35c), in which folded elongate minerals polygonize to an arcuate pattern consisting of smaller straight crystals. If post-kinematic recrystallization is exten-

kinematic, a random aggregate of grains of one mineral pseudomorphing a single crystal of another (chlorite after garnet). The chlorite is post-kinematic. From Spry (1969).

explained by the growth of a porphyroblast in a static situation over an existing schistosity. A later deformation may rotate an earlier porphyroblast or transpose S_e . Thus if S_i is not parallel to, or continuous with, S_e the porphyroblast is either pre-kinematic to D_2 , or syn-kinematic and deformation outlasted porphyroblast growth. It may be impossible to distinguish between these two possibilities.

Syn-kinematic mineral growth is probably the most common type in orogenic metamorphism because metamorphism and deformation are believed to occur generally in unison. It is also the most difficult to demonstrate unequivocally. A continuous schistosity, as illustrated in Figure 23, was probably generated by a process of dynamic (syn-kinematic) recrystallization. Aligned grains that are a mixture of bent crystals with undulose extinction and straight recrystallized grains support this conclusion more strongly, although they could also be a less likely combination of pre-kinematic and post-kinematic crystallization. It is probable, but largely a matter of faith, then, that syntectonic crystallization occurs during the development of a schistosity.

Misch (1969) described one clear indicator of definitively syn-kinematic mineral growth. Figure 36 is an

Figure 23-34 from John D. Winter (p. 528): Typical textures of pre-kinematic crystals (after Spry, 1969). Notice in (c) the classic **pressure shadow (or strain fringe)** where fibrous or granular minerals precipitate in the low-pressure sheltered zones adjacent to the porphyroblast, while the matrix foliation wraps around the rigid grain.

2. Pressure Solution Mass Transfer (Diffusive Mass Transfer)

The formation of pressure shadows is an integral part of the process of **pressure solution** (also called *diffusive mass transfer* or *wet diffusion*), which is one of the most prominent low-to-medium temperature deformation mechanisms in the crust (operating from ~150°C to 450°C):

1. **Dissolution:** Soluble minerals (primarily quartz and calcite) have chemical potentials that increase with compressive stress. Under elevated normal stress along grain boundaries, quartz or calcite dissolves into thin intergranular fluid films: $\mu = \mu_0 + \sigma_n \cdot V_m$ where μ is the chemical potential, σ_n is normal stress, and V_m is molar volume.
2. **Transport:** Dissolved chemical species diffuse through the fluid phase down the chemical potential gradient (away from high-stress surfaces toward regions of lower stress).
3. **Precipitation:** Ions reach the low-stress **pressure shadows** adjacent to rigid clasts (or extensional tension gashes), where the fluid is supersaturated relative to the local low stress, precipitating new mineral grains or fibrous aggregates.

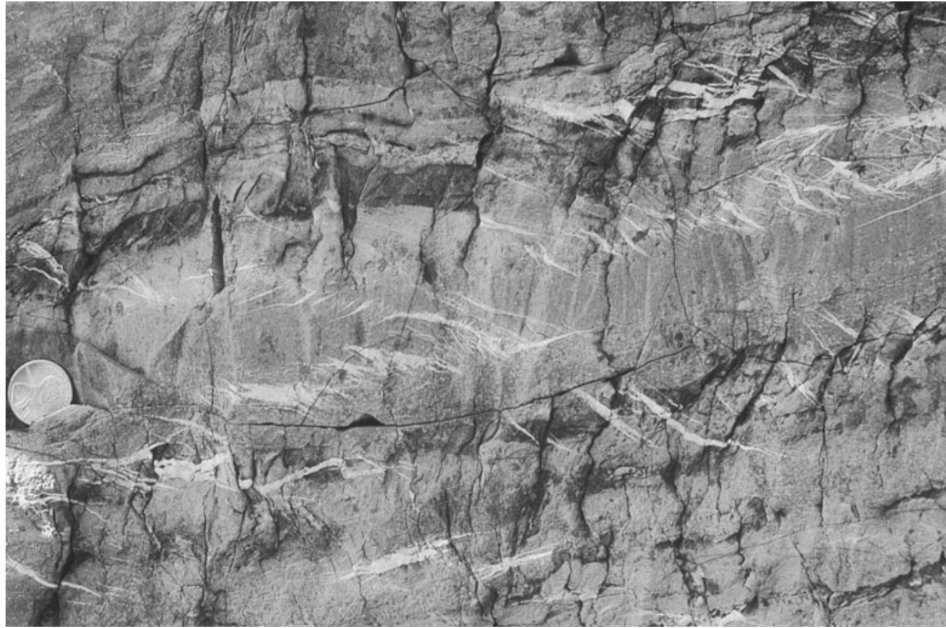


FIGURE 9.21 Bedding-perpendicular pressure solution seams (stylolites) and veins (white structures) in argillaceous limestone (Appalachians, Pennsylvania, USA). The middle bed is pure carbonate and does not contain as many seams. Note that the stylolites cut across bedding.

Figure 9-21 from Van der Pluijm & Marshak (p. 236): Complementary relationship between dissolution and precipitation during pressure solution. Material dissolved along compressive dissolution surfaces (stylolites) migrates and precipitates in regions of lowest stress (extensional veins and pressure shadows).

Fibrous Growth Fabrics in Pressure Shadows: Syntaxial vs. Antitaxial

Mineral fibers that precipitate in pressure shadows provide valuable kinematic records of the deformation history:

Syntaxial Fringes: Fibers grow outward from the rigid porphyroblast; oldest material is at the core clast.

Antitaxial Fringes: Fibers grow inward at the boundary with the porphyroblast; oldest material is at the outer tip.

Curved fibers in pressure shadows record progressive rotation of the incremental strain axes during non-coaxial (shear) deformation.



Official Exam Question Statement & In-Depth Critical Analysis

Directorate General of Mines Safety (DGMS) 2019 · Junior Scientific Officer Exam · Question 6

Question Statement

Q.6: When flattening occurs in rocks that contain rigid grains, zones of relatively low pressure develop on the sides of the crystals that are normal to the maximum compressive stress. These zones are called:

(a) Solution cleavage

(b) Pressure shadows

(c) Pressure solution

(d) Stylolites

Correct Answer: (b) Pressure shadows



These sheltered, low-stress zones that develop adjacent to rigid porphyroblasts during flattening are universally defined as pressure shadows (or strain shadows).

Exhaustive Option-by-Option Critical Analysis

(b) Pressure shadows

CORRECT

Why it is uniquely correct: In metamorphic petrology and structural geology (Winter, 2010, p. 527; Van der Pluijm & Marshak, 2004, pp. 236, 291; Ramsay & Huber, 1983; Passchier & Trouw, 2005), **pressure shadows** (also known as *strain shadows* or *pressure fringes*) are explicitly defined as the sheltered, low-mean-stress domains developed on the protected flanks of rigid crystals during tectonic flattening. Because the rigid clast bears the compressive load, the fluid in these protected areas has lower chemical potential, driving the precipitation of fibrous or granular quartz, calcite, or phyllosilicates.

(a) Solution cleavage

INCORRECT

Why it is incorrect: *Solution cleavage* (or pressure-solution cleavage / disjunctive cleavage) refers to a **spaced foliation fabric** formed by the dissolution and removal of soluble minerals, leaving behind planar seams concentrated in insoluble residues (clays, iron oxides, organic carbon). Solution cleavage is the planar dissolution fabric itself, not the localized low-pressure depositional domain adjacent to a rigid crystal.

(c) Pressure solution

INCORRECT

Why it is incorrect: *Pressure solution* is the **physical deformation mechanism** (the process of stress-induced dissolution, diffusion, and reprecipitation), whereas the question asks specifically for the name of the *zones* of relatively low pressure that develop adjacent to rigid grains. The process is pressure solution; the resulting low-pressure zone is the pressure shadow.

(d) Stylolites

INCORRECT

Why it is incorrect: *Stylolites* are jagged, serrated, tooth-like dissolution surfaces that form in regions of **maximum compressive stress** where extensive volume loss has occurred. Stylolites represent the exact physical opposite of pressure shadows: stylolites are the high-pressure dissolution source, whereas pressure shadows are the low-pressure precipitation sink!



Synthesis Reference Tables, Diagnostic Criteria & High-Yield Practice

Comparative Deformation Structures & 8 Exam-Grade Multiple-Choice Questions

Comparison of Pressure Solution Microstructures

Microstructure / Term	Stress Regime	Dominant Process	Typical Mineralogy / Composition	Diagnostic Field / Petrographic Form
Pressure Shadow (Strain Shadow)	Low Mean Stress (Sheltered Flank)	Precipitation of dissolved ions	Fibrous or granular Quartz, Calcite, Chlorite, Mica	Wings/fringes flanking rigid clasts (pyrite, garnet)
Stylolite	Maximum Compressive Stress (σ_1)	Stress-induced Dissolution (Volume Loss)	Insoluble residue: Illite, Smectite, Hematite, Bitumen	Serrated, tooth-like interpenetrating seams
Solution Cleavage	Maximum Compressive Stress (σ_1)	Pervasive Dissolution creating Spaced Fabric	Alternating Qtz/Cal microlithons & phyllosilicate cleavage domains	Parallel to anastomosing dark spaced cleavage seams
Extensional Vein (Tension Gash)	Least Principal Stress (σ_3 , Extension)	Fracture opening and mineral precipitation	Coarse fibrous or blocky Quartz, Calcite	Lenticular, en-échelon, or planar vein arrays

Kinematic Significance of Pressure Shadows

- **Foliation Wrapping:** Foliation planes (σ_1) bend smoothly around the rigid porphyroblast and its flanking pressure shadows, creating an eye-shaped (augen) geometry.
- **Sense of Shear Determination:** In ductile shear zones, asymmetric pressure shadows (such as σ -type and δ -type mantle/tail geometries) provide unequivocal indicators of the rotational sense of shear (dextral vs. sinistral).

- **Deformation Timing:** If pressure shadows contain unstrained fibers that truncate the matrix foliation, the fringe growth outlasted the main phase of foliation development (post-kinematic to syn-kinematic transition).

8 High-Yield Practice Questions on Pressure Shadows & Microstructures

1. In a deformed rock subjected to pure shear flattening, pressure shadows typically develop:

- (a) On faces of rigid grains perpendicular to the maximum compressive stress (σ_1)
- (b) Along the sides of rigid grains parallel to the extension direction (σ_3)
- (c) Inside the rigid grain along cleavage fractures
- (d) Exclusively along fault gouge seams

Answer: (b) — Pressure shadows develop in the low-pressure sheltered domains oriented parallel to the extension direction (σ_3) and perpendicular to σ_1 .

2. Which mechanism is primarily responsible for mass transfer during the formation of pressure shadows at low-to-medium metamorphic grades?

- (a) Dislocation glide
- (b) Pressure solution diffusion
- (c) Partial melting
- (d) Cataclastic flow

Answer: (b) Pressure solution diffusion — Dissolution at high-stress grain boundaries and diffusion through intergranular pore fluid toward low-stress sheltered shadows.

3. In an "antitaxial" pressure fringe, where is the youngest mineral growth located?

- (a) At the outer contact between the fringe and the rock matrix
- (b) At the inner contact directly adjoining the rigid porphyroblast clast
- (c) Uniformly distributed across the entire fringe
- (d) At the center of the porphyroblast

Answer: (b) — In antitaxial growth, new fibers precipitate at the boundary between the rigid core and the fringe, pushing older fiber segments outward.

4. Stylolites differ fundamentally from pressure shadows because stylolites represent sites of:

- (a) Maximum precipitation of quartz
- (b) Maximum stress-induced dissolution and volume loss
- (c) Magmatic crystallization
- (d) Brittle tensile fracture opening

Answer: (b) — Stylolites are jagged dissolution seams formed at points of highest compressive stress where material dissolves, whereas pressure shadows are sites of mineral precipitation in low-stress domains.

5. What causes the curved or sigmoidal shape of mineral fibers commonly observed in pressure shadows?

- (a) Rapid cooling of the metamorphic terrane
- (b) Progressive rotation of the rigid porphyroblast and principal strain axes during non-coaxial shear
- (c) Chemical reaction between quartz and feldspar
- (d) Sudden change in metamorphic fluid salinity

Answer: (b) — As non-coaxial (rotational) deformation proceeds, the incremental extension direction rotates relative to the clast, causing newly grown fiber segments to precipitate at changing angles, producing curved fibers.

6. Which mineral commonly forms prominent, easily recognizable cubic porphyroblasts with fibrous quartz pressure shadows in slates?

- (a) Halite
- (b) Pyrite
- (c) Gypsum
- (d) Biotite

Answer: (b) Pyrite — Rigid pyrite cubes (FeS_2) embedded in fine-grained slate matrices are textbook examples that develop spectacular quartz/chlorite pressure fringes.

7. In shear sense indicators, a rigid clast flanked by wedge-shaped tails of recrystallized material that are stair-stepped across the clast without a central embayment is termed a:

- (a) Δ -type clast
- (b) Σ -type clast
- (c) Stylolitic nodule
- (d) Mullion

Answer: (b) Σ -type clast — Sigma (Σ) porphyroclast systems have wedge-shaped tails that do not cross the reference median line, indicating non-coaxial shear.

8. Pressure solution creep becomes the dominant deformation mechanism under which typical crustal conditions?

- (a) Completely dry conditions at $> 900^\circ\text{C}$
- (b) Low-to-moderate temperatures (150°C – 400°C) in the presence of an intergranular fluid phase
- (c) High-strain-rate impact shock events
- (d) Temperatures below 0°C in permafrost

Answer: (b) — Pressure solution requires the presence of an intergranular aqueous fluid film and operates efficiently at low-to-moderate crustal temperatures (150°C – 400°C).